Keelan O'Doherty

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EDUCATION

University of Michigan Ann Arbor, MI · Aug 2026 – May 2028

Master of Science in Engineering (MSE), Aerospace Engineering, Space Systems

- Relevant Coursework: Spacecraft Technology, Space Policy & Management, Linear Systems Theory, Management of Space Systems Design, Astrodynamics

University of Alabama, Honors College Tuscaloosa, AL · Aug 2022 – May 2026

Bachelor of Science (BS) in Aerospace Engineering, Minor in Mathematics

- **Honors GPA 3.64/4.0** · Dean's List: every semester, Fall 2022 – Spring 2025
- Scholarships: Full-Ride Academic, Presidential, Engineering, Taylor Lewan Offensive Line Scholarship

Florence University of the Arts, Study Abroad Florence, Italy · May 2024

ENGINEERING EXPERIENCE

Northrop Grumman Sponsored Lunar Cargo Module (CAT-LIT) Ann Arbor, MI · Sep 2026 – Present

Systems Engineer, SPACE 582 Spacecraft Design (Northrop Grumman sponsored)

- Drafted 48 Level 0 and Level 1 requirements for a standardized lunar cargo module (8,000 kg cargo, 2,000 kg dry, 5/7/8 m variants) compatible with Starship and Blue Moon, each traced to a parent with a verification method and success criteria (NASA SP-2016-6105)
- Set the mission statement and six key performance parameters with threshold and objective values for the December mission design review and system requirements review
- Authored the cargo history and interface standards section of the lunar cargo delivery survey; checked lander and launch vehicle envelope and load data against the Falcon, Vulcan, Ariane 6, New Glenn and Starship user guides

University of Alabama Aerodynamics Research Laboratory (AAERO) Tuscaloosa, AL · Dec 2024 – May 2026

Undergraduate Research Assistant, Vertical Flight Society Chapter

- Designed, SLA 3D-printed, and bench-tested 4 UAV propeller geometries at 10,000 RPM to characterize thrust against acoustic performance
- Built the test setup and authored run procedures using load-cell thrust instrumentation and microphone acoustic profiling, structured for repeatable comparison across geometries
- Correlated measured results against Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEA) predictions to identify model-to-hardware divergence; presented findings at the URCA Conference 2025

Fixed-Wing eVTOL Aircraft - Design, Build & Flight Test Senior Capstone · Aug 2025 – May 2026

- Modeled the full ORCA airframe in SolidWorks from scratch for 3D printing, adapted from an existing VTOL design (fuselage, wings, spars, control surfaces, motor mounts), including a modular payload nose for optical, IR, and thermal sensor swaps
- Served as team pilot and ran the flight test campaign: phased hover and vertical-to-forward transition trials validating stability, control response, and endurance
- Modeled propulsion tilt-path dynamics and servo actuation sequencing in MATLAB/Simulink; validated the power budget against measured draw

Wilson & Company Phoenix, AZ · Jul 2024 – Aug 2024

Engineering Intern

- Gained hands-on experience with construction materials and techniques through 50+ hours of on-site exposure, documenting nonconformances and as-built conditions

ADDITIONAL EXPERIENCE

Watercraft Restoration - Owner & Technician Phoenix, AZ · Feb 2021 – Aug 2026

- Restored 65+ watercraft end to end, including 30+ full engine rebuilds and 30+ complete electrical rewires, from teardown through functional test
- Fabricated wire harnesses and mechanical subassemblies to torque specification; wrote repeatable teardown, build, and checkout procedures that reduced rework on later units

Pi Kappa Alpha Fraternity - Class President; Sports Chair Tuscaloosa, AL · 2022 – 2026

TECHNICAL PROJECTS

- **Embedded Vehicle Control System** - built Arduino-based control with GSM/IoT telemetry (SIM7000A) over LTE CAT-M1 for remote monitoring and actuation; designed power distribution using transistor logic and relay control, and wired the physical harness

TECHNICAL SKILLS

CAD: SolidWorks (three courses): bracket parts, assemblies and drawings for EcoPro equipment, cited in a published service bulletin; 90-part ORCA eVTOL airframe designed for 3D printing; 25+ personal product designs, printed and in use; own and run multiple FDM 3D printers

Mission & Orbit Analysis: STK 12 (sun-synchronous and mid-inclination coverage scenarios with rectangular and conic sensors, revisit and N-asset figures of merit; Astrogator lunar transfer with differential-corrector TLI and B-plane targeted LOI; low-thrust spiral with a custom propagator), patched-conic and Hohmann analysis in MATLAB

CFD & Simulation: ANSYS Fluent (axisymmetric density-based nozzle flow, SST k-omega, four-level mesh refinement to 137k cells; NACA 2412 with Spalart-Allmaras and SST; heated-channel convection), MATLAB/Simulink (DC-8 state-space pitch and roll-yaw control loops, quasi-1D nozzle solver, intercept simulation, solid and liquid stage sizing)

Programming & Tools: Python (NumPy, SciPy, Jupyter: laminate and buckling analysis, design searches), MATLAB, Arduino/C++, Git/GitHub, Excel, Microsoft 365

Additional: Languages: Spanish · Musical: Piano, Cello, Guitar